IF 451
Advance Web Programming

Meet5 - Introduction to MySQL Database

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Meet 5

Objective:

Students are able to implement various types of libraries and database to create web application using Node JS (C3)

OUTLINE

- *SQL vs NoSQL.*
- *Database – Collection – Document.*
- *MySQL Database Atlas account setup (cloud database)*

SQL vs NoSQL

What is SQL?

- SQL stands for Structured Query Language.
- It is used in relational databases (RDBMS) such as MySQL, PostgreSQL, Oracle, SQL Server.
- In SQL databases, data is stored in tables (rows and columns) with predefined schema (structure).
- You query the database using SQL commands like SELECT, INSERT, UPDATE, and DELETE.

What is NoSQL?

- NoSQL means Not Only SQL.
- It refers to databases that do not strictly use relational tables.
- Data can be stored in different formats, such as documents (JSON), key-value pairs, wide-column stores, or graphs.
- Common NoSQL databases: MongoDB, CouchDB, Cassandra, Redis, Neo4j.

Differences between SQL and NoSQL

Feature	SQL (Relational DB)	NoSQL (Non-relational DB)
Data structure	Tables (rows & columns)	JSON, key-value, documents, graph
Schema	Fixed schema (strict)	Schema-less (flexible)
Query language	SQL	Various (MongoDB uses JavaScript-like queries)
Scalability	Vertical (scale-up: bigger server)	Horizontal (scale-out: more servers)
Transaction support	Strong ACID (Atomic, Consistent, Isolated, Durable)	Often eventual consistency, but very scalable
Best for	Complex queries, structured data	Large, flexible, unstructured data

Advantages of SQL vs NoSQL

SQL (Relational DB)

- Strong data integrity
- Great for structured data with relationships (e.g., Students & Courses)
- Mature technology, widely supported
- Powerful query capabilities

NoSQL (Non-relational DB)

- Flexible schema → easy to store varied/unstructured data
- High scalability (good for big data & distributed systems)
- Often faster for simple lookups
- Naturally supports JSON-like storage (e.g., MongoDB)

JSON is SQL or NoSQL?

- Storing data in JSON format is usually considered **NoSQL**, because document-based databases (like MongoDB) use JSON (or BSON) as their native data model.
- However, modern SQL databases (like PostgreSQL and MySQL) also support JSON columns. That means you can still store JSON in SQL databases, but SQL itself is not naturally JSON-based.

Conclusion

- JSON storage → native in NoSQL.
- SQL → primarily tables, but can support JSON as a feature
- SQL → relational, structured, schema-based, best for strong consistency.
- NoSQL → flexible, scalable, JSON/document/key-value, best for unstructured or big data.
- JSON storage → belongs to NoSQL by nature, but some SQL databases can handle it too.

Database – Collection – Document

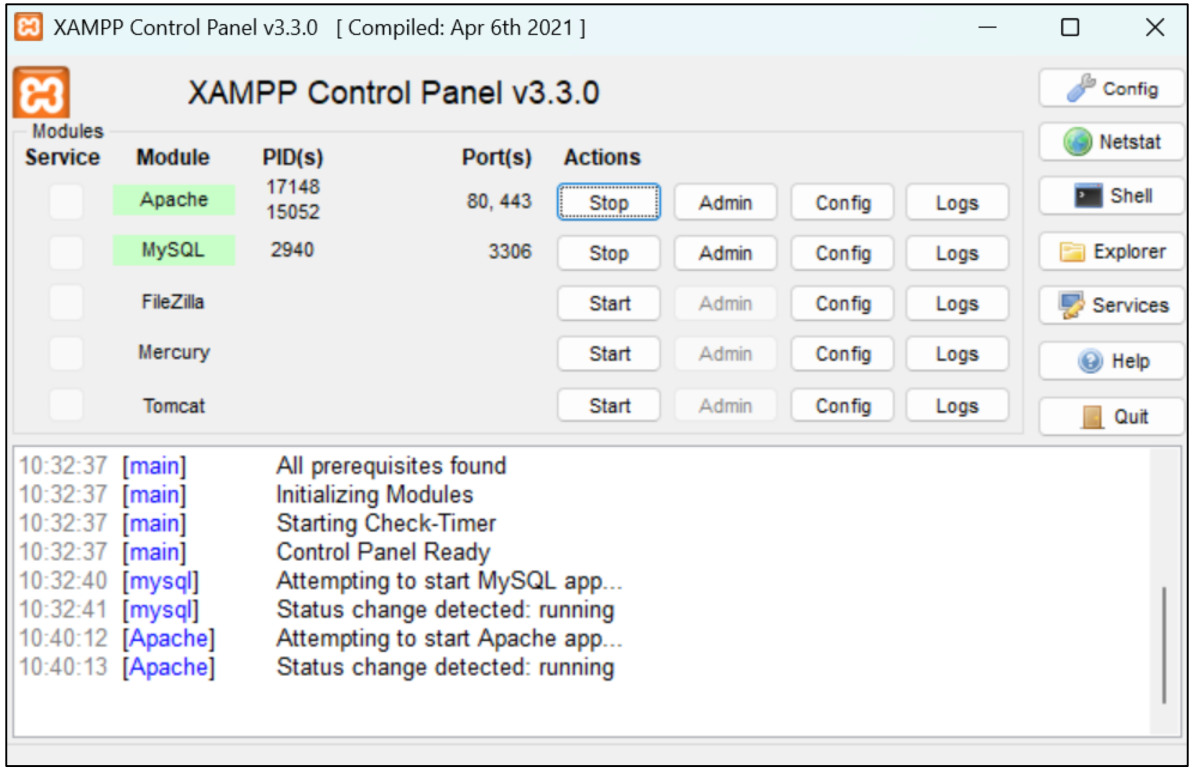
XAMPP

If you want to use mysql database, you can use it with install XAMPP to your computer. You can download it from official website XAMPP

https://www.apachefriends.org/download.html

Setup XAMPP

- To start mysql, after installation, you just only need to start MySQL.
- But if you want to use phpMyAdmin to support you in database management, you also need to start Apache.



phpMyAdmin vs MySQL CLI (Command Line Interface)

Advantages of Using phpMyAdmin

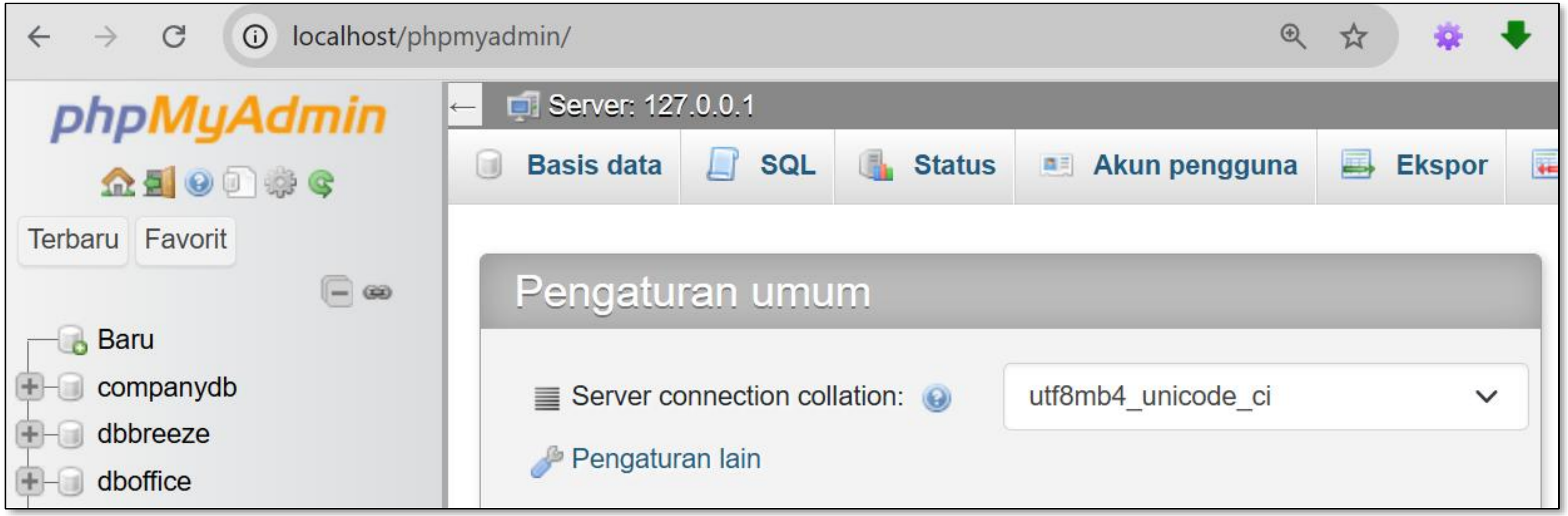
1. User-Friendly Interface
   - Provides a graphical web-based interface, so you don't need to memorize SQL commands.
   - Easier for beginners to manage databases, tables, and records.
2. Quick Data Visualization
   - You can instantly see tables, rows, and relationships without typing queries.
   - Great for checking whether data was inserted or updated correctly.
3. Convenient Features
   - Import/export tools (CSV, Excel, SQL dumps) are built-in.
   - Schema design tool (visual ER diagram) available.
4. Accessibility
   - Runs in a browser, so you can use it from different machines (if configured).

Advantages of Using MySQL CLI

1. Lightweight & Fast
   - No need for a web server or additional setup—just MySQL installed.
   - Faster for running queries if you're comfortable with SQL.
2. More Powerful & Flexible
   - Some advanced operations (e.g., scripting, batch queries, stored procedures) are easier in CLI.
   - Direct access to MySQL features without GUI limitations.
3. Secure & Low Resource Usage
   - No need to expose MySQL to a browser-based tool.
   - Less memory usage compared to phpMyAdmin running on Apache/Nginx.

Run phpMyAdmin

- After start apache and mysql in XAMPP,
- You could access your phpmyadmin via browser, with type "localhost/phpmyadmin" in url address



Create Database and Table with MySQL CLI

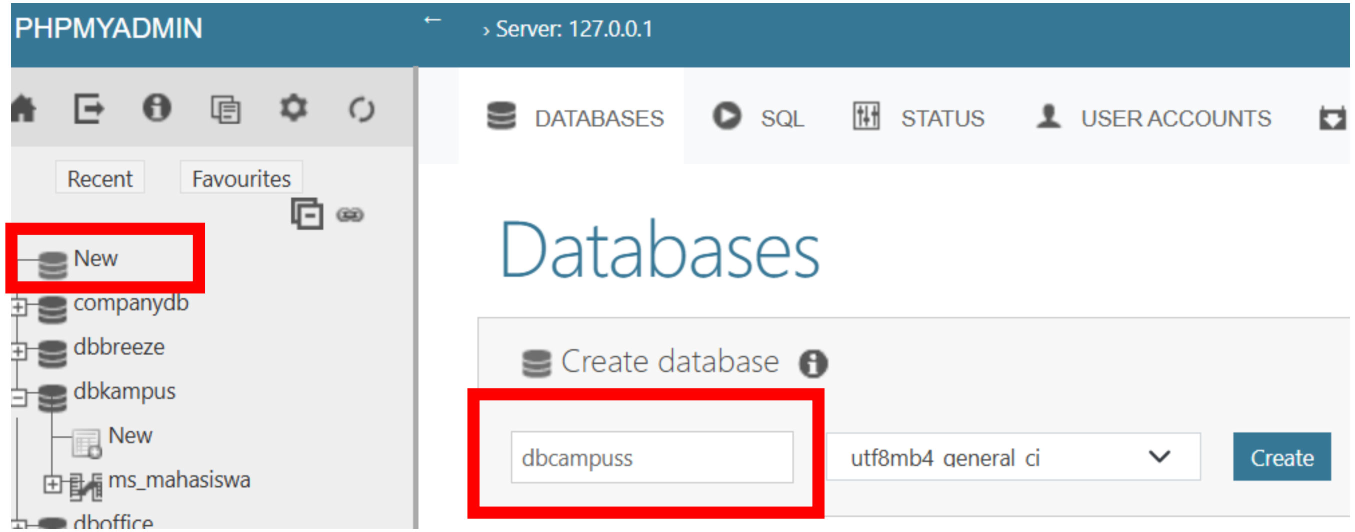
1. Open your terminal (command prompt), and change directory to your mysql installation folder, and type "**cd C:\xampp\mysql\bin**"
2. And to start using MySql CLI, type "**mysql -u root -p**", and enter your password (default password is empty – no password)
3. To create database, type "**CREATE DATABASE databasename;**"
4. To use your database, type "**USE databasename;**"
5. To create table, type:

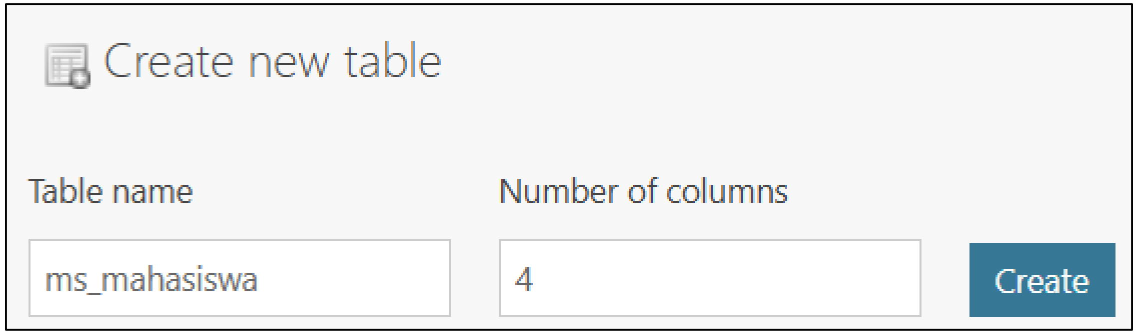
CREATE TABLE ms_mahasiswa (
nim VARCHAR(10) PRIMARY KEY,
nama VARCHAR(100),
jurusan VARCHAR(50),
alamat VARCHAR(200),
telepon VARCHAR(15)
);

```
C:\Users\user>cd C:\xampp\mysql\bin

C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or
Your MariaDB connection id is 103
Server version: 10.4.32-MariaDB mariadb.org binary distr

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab

Type 'help;' or '\h' for help. Type '\c' to clear the cu

MariaDB [(none)]> CREATE DATABASE dbkampus;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> use dbkampus
Database changed
MariaDB [dbkampus]> CREATE TABLE ms_mahasiswa (
    ->     nim VARCHAR(10) PRIMARY KEY,
    ->     nama VARCHAR(100),
    ->     jurusan VARCHAR(50),
    ->     alamat VARCHAR(200),
    ->     telepon VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.010 sec)
```

Create Database and Table with phpMyAdmin (1)

1. Visit phpmyadmin, using browser, and type "localhost/phpmyadmin"
2. To create database, choose "New" Database and give databasename in the textbox, and click button "create"
3. After database created, you can create table by type your tablename in the textbox Table name and choose how many column you want to create.





Create Database and Table with phpMyAdmin (2)

1. After table was created, you can setting table field. Create your **table field**, like the sample below.
2. To create Primary Key, you can select PRIMARY in the index column.
3. Click "save" button

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
nim Pick from Central Columns	VARCHAR	10	None			☐	PRIMARY PRIMARY
nama Pick from Central Columns	VARCHAR	100	None			☐	---
jurusan Pick from Central Columns	VARCHAR	50	None			☐	---
alamat Pick from Central Columns	VARCHAR	200	None			☐	---
telepon Pick from Central Columns	VARCHAR	15	None			☐	---

New data into your table (INSERT) - **C**reate

Via MySQL CLI

1. Open your terminal, and make sure you already "use databasename"
2. Write the SQL Query like the sample below

Via phpMyAdmin

1. Click SQL tab menu
2. Write the SQL Query like the sample below

```
INSERT INTO ms_mahasiswa (nim, nama, jurusan, alamat, telepon)

VALUES ('111','Wirawan','Informatika','Gading Serpong','0123456');
```

Show data (SELECT) - **R**ead

Via MySQL CLI

1. Open your terminal, and make sure you already "use databasename"
2. Write the SQL Query like the sample below

Via phpMyAdmin

1. Click SQL tab menu
2. Write the SQL Query like the sample below

```
SELECT * FROM ms_mahasiswa;
```

Edit data (UPDATE) - **U**pdate

Via MySQL CLI

1. Open your terminal, and make sure you already "use databasename"
2. Write the SQL Query like the sample below

Via phpMyAdmin

1. Click SQL tab menu
2. Write the SQL Query like the sample below

```
UPDATE ms_mahasiswa SET nama='Wirawan Istiono',
telepon='088123' WHERE nim='111‘;
```

Delete data (DELETE) - **D**elete

Via MySQL CLI

1. Open your terminal, and make sure you already "use databasename"
2. Write the SQL Query like the sample below

Via phpMyAdmin

1. Click SQL tab menu
2. Write the SQL Query like the sample below

```
DELETE FROM ms_mahasiswa WHERE nim='111';
```

MySQL Database Atlas account setup (cloud database)

Difference between local databases (like MySQL in XAMPP) and **cloud databases (hosted online):**

1. Location & Accessibility

- Local Database (XAMPP/MySQL):

  Installed on your personal computer or a local server. Only accessible on that machine (or within your local network if configured).

- Cloud Database:

  Hosted online by a provider (e.g., AWS RDS, PlanetScale, MongoDB Atlas). Accessible anywhere in the world via the internet.

2. Setup & Maintenance

- Local Database (XAMPP/MySQL):

  You install, configure, and maintain everything (XAMPP, MySQL, backups, updates).

- Cloud Database:

  The provider manages most of the setup, backups, scaling, and maintenance for you

Difference between local databases (like MySQL in XAMPP) and cloud databases (hosted online):

3. Performance & Scalability

- Local Database (XAMPP/MySQL):

  Limited to your machine's resources (CPU, RAM, disk space). Scaling up usually means buying a better computer/server

- Cloud Database:

  Can easily scale up or down (e.g., increase storage, RAM, or cluster size) without changing hardware.

4. Collaboration

- Local Database (XAMPP/MySQL):

  Hard to share. Other developers need access to your computer or you must export/import databases manually

- Cloud Database:

  Multiple users or applications can connect from different places at the same time.

Recommended Free Cloud MySQL Providers

Provider	Free Tier Details	Ideal For
Aiven (MySQL)	1 CPU, 1 GB RAM, 5 GB storage — indefinite free	Small projects & students
PlanetScale	5 GB storage, serverless scale, branching workflows	Modern dev workflow, prototypes
AWS RDS (12 mo.)	750 hrs/month, 20 GB storage, managed MySQL	Learners & temporary projects
FreeDB.tech	50 MB storage, lightweight, simple	Very small or test applications
Filess.io	10 MB storage, minimal connectivity	Tiny sample projects

NEXT WEEK'S OUTLINE

- Connecting Node.js to MySQL Database

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Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.

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Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.